DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot C

R.No: 15 Reg. No: 1925260144 Name: G Ganga Chandrika

Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-C15: Murder rate (per 100,000, 1976) [20 marks]

Brief: You are building a map story about Murder rate (per 100,000, 1976) across the United States for a local newspaper. Your comparison variable is Life expectancy (years, 1969-71). Your focus region is the North Central region. Your map uses 6 classes and the "magma" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$murder      # the variable you map
W <- d$life      # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of murder: ggplot(mapdat, aes(long, lat, group = group, fill = murder)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "magma") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 6 classes: equal interval, cut(V, 6), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 7)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of murders: count <- d$pop * 1000 * d$murder / 100000. Name the top 3 states by count and the top 3 states by murder [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the North Central region. Report the mean of murder in North Central and its ratio to the mean of all 50 states [1]; the highest state in North Central with its value [0.5]; and how many North Central states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a local newspaper [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

Code of Conduct: I certify that this work is my own. Signature: _____

Activity - 9

Map Storytelling

① Base Map

```
ggplot(mapdat, aes(long, lat, group=group, fill=murd
  geom_polygon(colour="White") +
  coord_map("albers", lat0=29.5, lat1=45.5)+
  scale_fill_viridis_c(option="magma") +
  labs(title="Murder Rate by State (1976)",
       fill="Murder Rate")
```

```
d$state[which.max(v)]
max(v)
d$state[which.min(v)]
min(v)
median(v)
```

② Regional story

```
tapply(v, d$zone, mean)
cor(v, w)
```

The highest & lowest regions are identified from their mean murder rates. Correlation shows the strength & direction of the relationship between murder rate and life expecta Correlation does not prove causation.

③ Classes change the story

```
equal <- cut(v, 6)
quantile <- cut(v, quantile(v, seq(0,1, length.out=7)),
                include.lowest = TRUE)
table(equal)
table(quantile)
```

equal intervals show extreme values more clearly. I prefer equal intervals for easy comparison.

④ Counts Versus Rates

```
count <- d$pop * 1000 * v/100000
head(sort(count, decreasing = TRUE), 3)
head(sort(v, decreasing = TRUE), 3)
```

Counts depend on population size & can mislead. Murder rates per 100,000 people are better for comparision.

⑤ North central story

```
nc <- d$zone == "North central"
mean(v[nc])
mean(v[nc])/mean(v)
d$state[nc][which.max(v[nc])]
sum(v[nc] > median(v))
```

Headline: North central states show varied Murder rates.

Evidence: Compare the regional mean with the national mean & identify the highest-rate state.

Recommendation: Use a colour-coded murder-rate map with a clear legend.